

K3 SURFACES EXAM

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COMPLEX CURVES ON A K3 SURFACE

Exercise 0.1 (2.6a). Let C be a smooth genus g curve which can be embedded in a K3 surface M , and X the family of all deformations of C in M . Prove that $\dim X \leq g$.

Proof. We are asked to compute $\dim X = \dim |C| = h^0(\mathcal{O}_S(C)) - 1$.

By definition of Euler characteristic,

$$\chi(\mathcal{O}(C)) = h^0(\mathcal{O}(C)) - h^1(\mathcal{O}(C)) + h^2(\mathcal{O}(C)).$$

To compute this Euler characteristic we use Riemann-Roch theorem for curves in surfaces:

$$\chi(\mathcal{O}(C)) = \chi(\mathcal{O}_S) + \frac{1}{2}(\mathcal{O}(C), \mathcal{O}(C) - K_S) = 2 + \frac{1}{2}\mathcal{O}(C)^2$$

since S is a K3 surface.

To compute the self-intersection number we use genus formula:

$$(\mathcal{O}(C), \mathcal{O}(C) + K_S) = \mathcal{O}(C)^2 = 2g - 2$$

since S is a K3 surface.

Thus $\chi(\mathcal{O}(C)) = g + 1$ and

$$g + 1 = h^0(\mathcal{O}(C)) - h^1(\mathcal{O}(C)) + h^2(\mathcal{O}(C)).$$

By Serre duality and S being a K3, we obtain two equations:

$$h^2(\mathcal{O}(C)) = h^0(\mathcal{O}(-C)), \quad h^1(\mathcal{O}(C)) = h^1(\mathcal{O}(-C)).$$

The ideal sheaf $\mathcal{O}(-C)$ exact sequence gives the long exact sequence

$$0 \rightarrow H^0(\mathcal{O}(-C)) \hookrightarrow H^0(\mathcal{O}_S) = \mathbb{C} \rightarrow H^0(\mathcal{O}_C) = \mathbb{C} \rightarrow H^1(\mathcal{O}(-C)) \rightarrow \dots$$

First note that $h^0(-C) = 0$ since the global sections of the ideal sheaf $\mathcal{O}(-C)$ are global holomorphic functions on S vanishing along C —only the constant zero function lies here. Thus $H^0(\mathcal{O}_S) = \mathbb{C} \rightarrow H^0(\mathcal{O}_C) = \mathbb{C}$ in the long exact sequence is an injection, and thus surjective, making $H^1(\mathcal{O}(-C)) = 0$. Thus $\dim X = g$. $\square$

Exercise 0.2 (2.7). Let C be a singular, irreducible complex curve on a K3 surface X . Prove that $(C, C) \geq 0$.

Proof. This is an application of the genus formula. Since $K_X = \mathcal{O}_X$ it suffices to show that the arithmetic genus $p_a(C)$ is not zero. This follows from the fact that C is singular, since its arithmetic genus, defined as $p_a(C) := 1 - \chi(\mathcal{O}_C)$, satisfies

$$(0.2.1) \quad p_a(C) = g(C) + h^0(\delta)$$

where $g(C)$ is the geometric genus of C , defined as the genus of the normalization $\tilde{C}$, i.e. $g(C) := g(\tilde{C})$, while $\delta := \nu_* \mathcal{O}_{\tilde{C}} / \mathcal{O}_C$ where $\nu : \tilde{C} \rightarrow C$ is the normalization map.

To prove Equation 0.2.1 first note that $p_a(C) = h^1(\mathcal{O}_C)$ and $g(C) = h^1(\mathcal{O}_{\tilde{C}})$, where the latter uses Serre duality since normalization of a curve is smooth.

Now use the normalization long exact sequence in cohomology

$$0 \rightarrow H^0(\mathcal{O}_C) \rightarrow H^0(\nu_* \mathcal{O}_{\tilde{C}}) \rightarrow H^0(\delta) \rightarrow H^1(\mathcal{O}_C) \rightarrow H^1(\nu_* \mathcal{O}_{\tilde{C}}) \rightarrow H^1(\delta)$$

Note the following three facts: the map $H^0(\mathcal{O}_C) \rightarrow H^0(\nu_* \mathcal{O}_{\tilde{C}})$ is an isomorphism, $H^1(\nu_* \mathcal{O}_{\tilde{C}}) \cong H^1(\mathcal{O}_{\tilde{C}})$ and $H^1(\delta) = 0$. The first is due to both sheaves' global sections being constant functions, the second by ν being finite (Lecture 9), and the third because δ is supported in finitely many points. The latter is because the support of δ are the singularities of C (which must be finitely many by compactness): the stalk δ_p is the quotient of the integral closure of the stalk $\mathcal{O}_p$ by $\mathcal{O}_p$, which is nonzero if and only if p is not a smooth point. Thus we have the short exact sequence

$$0 \longrightarrow H^0(\delta) \longrightarrow H^1(\mathcal{O}_C) \longrightarrow H^1(\nu_* \mathcal{O}_{\tilde{C}}) \longrightarrow 0$$

yielding Equation 0.2.1.

Finally, since C is singular δ is nonzero and thus $h^0(\delta) > 0$. $\square$

Lemma 0.3 (Genus formula). *Let C be any curve on a surface S . Then*

$$(0.3.1) \quad 2p_a(C) - 2 = (C.C + K_S).$$

Proof. By Remark 0.4 below we can take Euler characteristic of the ideal sheaf exact sequence of C to obtain

$$\chi(\mathcal{O}(-C)) + \chi(\mathcal{O}_C) = \chi(\mathcal{O}_S).$$

Since $p_a(C) = h^0(K_S) = 1 - \chi(\mathcal{O}_C)$, we obtain that $p_a(C) = 1 - \chi(\mathcal{O}_S) - \chi(\mathcal{O}(-C))$. To compute $\chi(\mathcal{O}(-C))$ we use Riemann-Roch formula for curves on surfaces to get

$$\chi(\mathcal{O}(-C)) = \chi(\mathcal{O}_S) + \frac{(\mathcal{O}(-C).C - K_S)}{2}$$

Taking duals on both entries of the intersection form and substituting in our previous expression for $p_a(C)$ we obtain the result. $\square$

Remark 0.4. To prove that Euler characteristic is additive, i.e. that for an exact sequence

$$0 \longrightarrow E \longrightarrow F \longrightarrow G \longrightarrow 0$$

we must have

$$\chi(F) = \chi(E) + \chi(G),$$

we take the long exact sequence in cohomology and use the general fact that for a finite exact sequence of vector spaces

$$0 \longrightarrow V_0 \longrightarrow V_1 \longrightarrow \cdots \longrightarrow V_n \longrightarrow 0$$

we must have

$$\sum_i (-1)^i \dim V_i,$$

which basically follows by dimension theorem.